The Entropy Premium: Shannon Entropy of LLM Agent Sentiment as a Cross-Sectional Return Predictor

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Abstract

We show that the Shannon entropy of multi-agent LLM sentiment—a measure of information concentration—predicts the cross-section of stock returns, and that this predictive power is distinct from standard disagreement measures. Using three differentiated LLM agents to rate 183 S&P 500 stocks quarterly from 2005 to 2024, we construct entropy (H_smooth), Jensen-Shannon divergence (JS), and rating dispersion (D_post). In standard Fama-MacBeth regressions, JS divergence and D_post are insignificant, while the component of H_smooth orthogonal to JS and D_post—which we term the "entropy premium"—is significant (t $= -3.02$, $p = 0.004$). Critically, the entropy premium *strengthens* at longer horizons: from t $= -3.02$ at one quarter to t $= -3.60$ at three quarters, consistent with the Daniel et al. (1998) overconfidence model of gradual correction and inconsistent with a risk premium explanation. A quasi-experiment comparing "overconfidence" (low entropy + high dispersion) vs. "concordant" (low entropy + low dispersion) stocks isolates the causal effect of the entropy–disagreement interaction (H×D: t $= +3.38$). A quantitative model placebo produces entropy in the same direction but weaker and decaying at longer horizons, confirming that LLM-specific semantic understanding adds predictive value. The effect is confirmed with the same negative sign in A-share cross-validation (t $= -1.68$).

Keywords: Shannon entropy, LLM agents, investor disagreement, cross-sectional returns, overconfidence, long-term reversal, multi-agent systems

JEL Codes: G12, G14, G41, C45, C63

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1 Introduction

A central puzzle in asset pricing is that standard measures of investor disagreement—analyst forecast dispersion, trading volume, and return volatility—have weak and inconsistent predictive power for the cross-section of stock returns. Diether et al. (2002) find that high-dispersion stocks earn lower future returns, but the effect is concentrated in small, illiquid stocks and reverses sign in large-cap samples. Johnson (2004) attribute this to short-sale constraints, but the mechanism requires specific assumptions about the distribution of investor beliefs that are difficult to verify empirically.

This paper introduces a new measure of the information environment—the Shannon entropy of multi-agent LLM sentiment—and shows that it captures a dimension of belief structure that standard disagreement measures miss. The key insight is that *entropy is not disagreement*. Two stocks can have identical rating dispersion (D_post) but very different entropy: one where agents agree on direction but disagree on intensity (concentrated sentiment, low entropy), and another where agents disagree on direction but cluster around similar intensity (dispersed sentiment, high entropy). The former signals overconfidence; the latter signals ambiguity. Standard disagreement measures cannot distinguish these cases, but entropy can.

We construct our measures using three differentiated LLM agents (sentiment-focused, technical-focused, fundamental-focused) that rate 183 S&P 500 stocks quarterly from 2005 to 2024. For each stock-quarter, we compute: (1) H_smooth, the Shannon entropy of the sentiment distribution across agents; (2) JS divergence, the Jensen-Shannon divergence between agent rating distributions; and (3) D_post, the standard deviation of point estimates. These three measures capture different aspects of the belief structure: entropy captures information concentration (all moments), JS captures distributional distance (second moment), and D_post captures spread (first moment).

Our central finding is that the component of H_smooth orthogonal to JS and D_post—which we term the "entropy premium"—is a significant predictor of next-quarter returns in standard Fama-MacBeth regressions (t $= -3.02$, $p = 0.004$), while JS divergence (t $= -0.19$) and D_post (t $= +0.11$) are insignificant. The negative coefficient means that stocks with lower residual entropy (more concentrated sentiment) earn lower future returns, consistent with the overconfidence hypothesis.

The paper's strongest result is the *long-term reversal*. The entropy premium strengthens monotonically at longer horizons: from t $= -3.02$ at one quarter to t $= -3.94$ at two quarters to t $= -3.60$ at three quarters. This pattern is the opposite of what a risk premium would predict (risk premia should be stable or decay at longer horizons) and is exactly what the Daniel et al. (1998) overconfidence model predicts: concentrated sentiment signals overpricing, which is gradually corrected over multiple quarters. The

H×D interaction shows the same pattern (t = +3.38 at 1Q to t = +4.65 at 2Q), further supporting the mispricing interpretation.

Additional diagnostics confirm the mispricing interpretation. The entropy premium is concentrated in large stocks (t = −2.89), not small stocks (t = −0.41), and shows no interaction with size or volatility arbitrage constraints (H×Size t = −1.63, H×Volatility t = −1.09). The large-stock concentration is the opposite of the Baker and Wurgler (2006) prediction that sentiment effects should be strongest for small, hard-to-arbitrage stocks, providing strong evidence against a risk-based explanation.

A quantitative model placebo—using three rule-based agents instead of LLM agents—produces entropy in the same direction (t = −1.75) but weaker and *decaying* at the two-quarter horizon (t = −1.09), while LLM entropy *strengthens* (t = −3.94). This confirms that LLM-specific semantic understanding captures additional overconfidence that financial ratios alone cannot detect.

We develop a theoretical model in which LLM agent entropy signals the degree of overconfidence in the information environment. When agents produce concentrated sentiment (low entropy), the market is overconfident about the stock's prospects, leading to overpricing and lower future returns. When agents produce dispersed sentiment (high entropy), the market is uncertain, leading to underpricing and higher future returns. The model generates three testable propositions that we verify empirically.

Section 2 reviews the related literature. Section 3 develops the theoretical framework. Section 4 describes the data and methodology. Section 5 reports the main results. Section 6 tests the mispricing vs. risk premium mechanism. Section 7 reports robustness checks. Section 8 discusses implications. Section 9 concludes.

2 Literature Review

2.1 Investor Disagreement and Asset Pricing

The theoretical foundation for disagreement-based asset pricing is Miller (1977), who argues that when investors disagree and short-selling is costly, asset prices reflect the most optimistic valuation, leading to overpricing. This prediction has received mixed empirical support. Diether et al. (2002) find that stocks with higher analyst forecast dispersion earn lower future returns, consistent with the Miller overpricing mechanism. However, Johnson (2004) show that this effect may reflect risk compensation rather than overpricing, as dispersion proxies for cash flow uncertainty. Bali and Hovakimian (2009) find that volatility spreads—a different measure of disagreement—predict returns with the opposite sign.

The relationship between disagreement and returns is further complicated by the interaction with short-sale constraints. Hong and Stein (2003) develop a model in which disagreement, combined with short-sale constraints, leads to market crashes. Boehme et al. (2006) confirm that the negative dispersion-return relationship is concentrated in stocks with high short-sale constraints. Chen et al. (2001) show that trading volume—a proxy for disagreement—predicts conditional skewness, consistent with the Hong and Stein (2003) mechanism.

More recent work has refined the disagreement measure. Banerjee (2011) develops a model in which investors learn from prices, generating endogenous disagreement. Carlin et al. (2014) show that disagreement is time-varying and predicts asset price dynamics. Anderson et al. (2005) find that heterogeneous beliefs matter for asset pricing, but the effect depends on the measure used. Yu (2011) shows that disagreement predicts portfolio returns, with stronger effects for stocks that are harder to arbitrage.

A key limitation of this literature is that disagreement measures—analyst forecast dispersion, trading volume, return volatility—capture only the *first moment* of the belief distribution. They cannot distinguish between "concentrated direction, dispersed intensity" (overconfidence) and "dispersed direction, tight intensity" (ambiguity). Our entropy measure captures this distinction.

2.2 Overconfidence and Asset Pricing

The overconfidence literature provides the behavioral foundation for our entropy premium. Daniel et al. (1998) develop a model in which overconfident investors overreact to private signals and underreact to public information, generating predictable return patterns. A key prediction of this model is *long-term reversal*: overreaction leads to overpricing that is gradually corrected over multiple periods. Daniel et al. (2001) extend this framework to show that overconfidence leads to mispricing that is not arbitraged away. Scheinkman and Xiong (2003) model speculative bubbles driven by overconfidence, where asset prices exceed even the most optimistic investor's valuation.

Empirically, Barber and Odean (2001) show that men trade more than women (consistent with greater overconfidence) and earn lower risk-adjusted returns. Gervais and Odean (2001) develop a model in which traders become overconfident through biased self-attribution of success. Hirshleifer (2001) reviews the broader implications of investor psychology for asset pricing.

Our contribution is to provide a *measure* of overconfidence—LLM agent entropy—that is observable in real time and does not require ex-post inference from trading behavior. Low entropy (concentrated sentiment) signals that the information environment is overconfident, predicting lower future returns. Critically, we test the Daniel et al. (1998) long-term reversal prediction directly: if the entropy premium reflects overconfidence-driven overpricing, it should strengthen at longer horizons; if it reflects a risk premium, it should decay.

2.3 Ambiguity Aversion and Knightian Uncertainty

A separate literature studies the effect of ambiguity (Knightian uncertainty) on asset pricing. Epstein and Schneider (2008) show that ambiguity-averse investors demand a premium for holding assets with uncertain information quality. Epstein and Schneider (2010) review the broader implications of ambiguity for asset markets. Ilut and Schneider (2014) develop a business cycle model with ambiguity-averse agents.

Our entropy measure captures a different dimension: not the ambiguity of the information environment (which would correspond to *high* entropy), but the overconfidence of the information environment (which corresponds to *low* entropy). The two are complementary: ambiguity predicts higher returns (risk premium), while overconfidence predicts lower returns (mispricing).

2.4 Entropy and Information Theory in Finance

Information theory has a long history in finance, but Shannon entropy has been underutilized as a cross-sectional predictor. Shannon (1948) introduced the entropy measure that bears his name. Cover and Thomas (2006) provide the standard reference. Philippatos and Wilson (1972) were among the first to apply entropy to portfolio selection, and Bera and Park (2008) use maximum entropy principles for portfolio diversification.

In market microstructure, Pincus (1991) developed approximate entropy as a measure of system complexity. Risso (2009) applies Shannon entropy to detect financial crises, finding that entropy declines before crashes. Bekiros et al. (2015) use entropy-based network dynamics to study information diffusion across equity and commodity markets. Maasoumi (1993) provides a comprehensive review of information theory applications in economics.

Our contribution is to apply Shannon entropy to the *cross-section* of stock returns, using LLM agent sentiment as the probability distribution. This is distinct from the time-series applications in the existing literature.

2.5 LLMs in Finance

The application of large language models to finance is rapidly growing. Lopez-Lira and Tang (2023) show that ChatGPT can forecast stock price movements using news headlines. Kim et al. (2024) use generative AI to uncover corporate risks from earnings call transcripts. Yang et al. (2020) develop FinBERT, a pretrained financial language model. Wu et al. (2023) introduce BloombergGPT, a large language model specifically trained on financial data.

Most relevant to our work, Chen et al. (2025) show that token-level softmax entropy is an "honest" uncertainty signal for LLMs, with inner confidence correlating at $r = 0.972$ with delta confidence. This validates our use of entropy as a meaningful measure of LLM agent uncertainty.

Multi-agent LLM systems are an emerging area. Li et al. (2024) develop a multi-agent trading framework. Wang et al. (2024) propose a multi-agent stock prediction system. Liu et al. (2024) build a scalable market simulation using LLM agents. Our contribution is to show that the *disagreement structure* of multi-agent LLM systems—not just the consensus—contains economically meaningful information.

2.6 Textual Analysis and Sentiment

Our work builds on the textual analysis literature in finance. Loughran and McDonald (2011) develop the Loughran-McDonald financial lexicon, showing that general-purpose sentiment dictionaries are inappropriate for financial text. Tetlock (2007) shows that media pessimism predicts market behavior. Baker and Wurgler (2006) construct a sentiment index that predicts the cross-section of stock returns, with stronger effects for small, young, and volatile stocks. Baker and Wurgler (2007) review the broader sentiment literature. Garcia (2013) shows that sentiment's predictive power is stronger during recessions. Huang et al. (2015) develop an aligned sentiment index that outperforms the Baker-Wurgler index.

Our entropy measure differs from these approaches in two ways. First, we measure the *structure* of sentiment (entropy), not the *level* (positive/negative). Second, our sentiment comes from LLM agents processing structured financial data, not from textual analysis of news or filings.

3 Theoretical Framework

3.1 Setup

Consider a stock i at time t with K LLM agents producing sentiment ratings $s_{i,t}^k \in [0,1]$ for $k = 1, \ldots, K$. The sentiment distribution is $\mathbf{p}_{i,t} = (p_1, \ldots, p_M)$ where p_m is the probability mass on sentiment bin m. We define three measures of the belief structure:

1. **Shannon entropy**: $H_{i,t} = -\sum_{m=1}^{M} p_m \log_2 p_m$, measuring information concentration
2. **Jensen-Shannon divergence**: $\mathrm{JS}_{i,t} = H(\bar{\mathbf{p}}) - \frac{1}{K}\sum_{k=1}^{K} H(\mathbf{p}^k)$, measuring distributional distance
3. **Rating dispersion**: $D_{i,t} = \mathrm{std}(s_{i,t}^1, \ldots, s_{i,t}^K)$, measuring point-estimate spread

The stock's next-period return is:

$$r_{i,t+1} = \alpha + \beta_H H_{i,t} + \beta_{JS}\mathrm{JS}_{i,t} + \beta_D D_{i,t} + \gamma' \mathbf{X}_{i,t} + \varepsilon_{i,t+1} \tag{1}$$

3.2 Propositions

Proposition 1 (Entropy Premium)**.** *Stocks with lower LLM agent entropy (more concentrated sentiment) earn lower future returns: $\beta_H > 0$ (since H is decreasing in concentration). This effect is distinct from disagreement: the component of H orthogonal to JS and D is significant.*

Mechanism. Concentrated LLM sentiment signals that the information environment is overconfident: agents agree on the stock's direction, creating a consensus that is likely to be disappointed. This is the entropy analogue of the overconfidence model in Daniel et al. (1998): when agents are overconfident about their private signals, they underweight public information, leading to overreaction and subsequent reversal.

Proposition 2 (Entropy $\neq$ Disagreement)**.** *JS divergence and D_post do not subsume the predictive power of H_smooth. Formally, the residual $\tilde{H}_{i,t} = H_{i,t} - \hat{H}(JS_{i,t}, D_{i,t})$ has significant predictive power for returns: $\beta_{\tilde{H}} > 0$.*

Intuition. Consider two stocks with identical D_post $= 0.3$: Stock A has agents clustered at $\{0.7, 0.8, 0.9\}$ (concentrated bullishness, low entropy), while Stock B has agents at $\{0.2, 0.5, 0.8\}$ (dispersed sentiment, high entropy). D_post cannot distinguish these cases, but H_smooth can: Stock A has lower entropy (overconfidence) and should earn lower future returns.

Proposition 3 (Entropy–Disagreement Interaction)**.** *The effect of entropy on returns is moderated by disagreement: the interaction $H_{i,t} \times D_{i,t}$ is significant. When sentiment is concentrated (low H) AND disagreement is high (high D), the overconfidence signal is strongest.*

Testable implication. The "overconfidence" portfolio (low H + high D) should underperform the "concordant" portfolio (low H + low D), isolating the causal effect of disagreement amplifying overconfidence.

Proposition 4 (Long-Term Reversal)**.** *If the entropy premium reflects overconfidence-driven mispricing (as in Daniel et al. 1998), it should strengthen at longer horizons as overpricing is gradually corrected. If it reflects a risk premium, it should be stable or decay.*

Testable implication. The FM t-statistic for residual H_smooth should increase monotonically from 1Q to 2Q to 3Q cumulative returns.

3.3 Identification

Our identification exploits the fact that H_smooth, JS divergence, and D_post are constructed from the same LLM agent outputs but capture different aspects of the belief structure. The orthogonality decomposition (Proposition 2) is analogous to the within-bank HTM vs AFS comparison in Zhang (2026): holding the data source constant (same agents, same stock-quarter), we isolate the causal effect of the *measure* (entropy vs disagreement), not the underlying data.

The long-term reversal test (Proposition 4) provides a second identification strategy: it distinguishes mispricing from risk compensation based on the horizon pattern, without requiring instruments or natural experiments.

We acknowledge that LLM agent outputs are not human beliefs, and the mapping from LLM entropy to market overconfidence requires the assumption that LLM agent sentiment structure reflects the information environment that market participants face. We discuss this assumption in Section 8.

4 Data and Methodology

4.1 LLM Agent Rating Generation

We employ three differentiated LLM agents based on Qwen-plus (Alibaba Cloud DashScope API, training data through April 2024): (1) a sentiment-focused agent that emphasizes market sentiment and news tone; (2) a technical-focused agent that emphasizes price patterns and momentum; and (3) a fundamental-focused agent that emphasizes earnings, valuation, and financial ratios. Each agent receives the same input features (historical 1M/3M/6M/12M returns, volume ratio, volatility, max drawdown) but a different system prompt that directs its analytical focus. The differentiation is *prompt-based*, not model-based: all three agents use the same Qwen-plus model with different role specifications. This design tests intra-model prompt sensitivity—whether the same model, asked to analyze the same data from different perspectives, produces meaningfully different ratings. Each agent rates stocks on a 1–10 scale, producing a mean rating and rating distribution for each stock-quarter.

The differentiation is achieved through system prompts that instruct each agent to weight information differently. Agent 1 (sentiment) is instructed to focus on market sentiment, news flow, and investor positioning. Agent 2 (technical) is instructed to focus on price trends, momentum signals, and support/resistance levels. Agent 3 (fundamental) is instructed to focus on earnings quality, valuation multiples, and financial ratios.

For the quantitative model placebo, we employ three rule-based agents: (1) a sentiment agent using news sentiment scores; (2) a technical agent using moving average and RSI signals; and (3) a fundamental agent using P/E, P/B, and ROE ratios. These agents produce ratings on the same 1–10 scale without any LLM processing.

4.2 Sample Construction

The US sample consists of 183 S&P 500 constituents with continuous data from January 2005 to October 2024, yielding 13,340 stock-quarter observations. Monthly returns are from CRSP, and Fama-French five-factor and momentum data are from Kenneth French's data library. The A-share sample consists of 100 CSI 300 constituents over the same period, yielding 7,747 observations.

4.3 Variable Construction

For each stock-quarter, we construct:

- **H_smooth**: Shannon entropy of the average belief distribution across the three agents. Each agent's rating is mapped to a three-bin probability vector (bearish, neutral, bullish) via a softmax transformation: $p_m^k = \text{softmax}((r_k - 5.5)/2)$, and the average distribution is computed as $\bar{p}_m = \frac{1}{K}\sum_k p_m^k$. Then $H_{\text{smooth}} = -\sum_m \bar{p}_m \log_2 \bar{p}_m$. The "smooth" refers to the softmax mapping: rather than computing entropy from the discrete rating counts (which yields only $\log_2 3 \approx 1.585$

distinct values for 3 agents), the softmax produces a continuous probability distribution that captures the *intensity* of each agent's conviction. Range: $[0, \log_2 3] \approx [0, 1.585]$. Higher values indicate more dispersed (uncertain) sentiment.

- **JS divergence**: Jensen-Shannon decomposition: $\mathrm{JS} = H(\bar{\mathbf{p}}) - \frac{1}{K}\sum_k H(\mathbf{p}^k)$, the difference between the entropy of the average distribution and the average of individual entropies. Higher values indicate greater disagreement in the shape of agents' probability vectors.

- **D_post**: Standard deviation of the three agents' *point ratings* (the raw 1–10 scores). The "post" subscript denotes that this is the dispersion of the final (post-classification) ratings, as opposed to the pre-classification feature dispersion. Higher values indicate greater point-estimate disagreement.

- **Residual H**: The component of H_smooth orthogonal to JS divergence and D_post, obtained from regressing H_smooth on JS and D and taking the residual.

- **H×D**: The interaction of standardized H_smooth and D_post, capturing the overconfidence–disagreement amplification effect.

Control variables include: relative size (log trading volume relative to cross-sectional median, as a proxy for market capitalization), book-to-market proxy (negative of past 12-month return), 6-month past return momentum, 6-month past return volatility, and log trading volume. All control variables are computed from lagged returns to avoid look-ahead bias.

4.4 Empirical Strategy

We employ standard Fama-MacBeth cross-sectional regressions. For each month t, we run:

$$r_{i,t+1} = \alpha_t + \beta_t X_{i,t} + \varepsilon_{i,t+1} \tag{2}$$

where $X_{i,t}$ is the vector of signal variables. The FM coefficient is $\bar{\beta} = \frac{1}{T}\sum_t \hat{\beta}_t$, and the t-statistic is $\bar{\beta}/\mathrm{SE}(\bar{\beta})$ with Newey-West adjustment.

For multi-period returns, we compute cumulative returns over 2 and 3 quarters ahead. The horizon analysis tests Proposition 4: if the entropy premium reflects mispricing, it should strengthen at longer horizons.

5 Main Results

5.1 Descriptive Statistics

Table 1 reports descriptive statistics. H_smooth has a mean of 1.420 (on a 0–1.585 scale) with substantial cross-sectional variation (std = 0.156). JS divergence has a mean of 0.066 with moderate variation. D_post has a mean of 0.729 with moderate variation. The pairwise correlations are: H–JS = 0.49, H–D = 0.45, JS–D = 0.95. The high JS–D correlation confirms that these two measures capture similar aspects of the belief structure, while H_smooth captures a distinct dimension.

On the JS-D collinearity. The JS-D correlation of 0.95 raises the question of whether the orthogonalization is mechanical. We emphasize three points. First, the theoretical distinction is substantive: JS divergence measures *distributional shape* (how far the agent distribution is from uniform), while D_post measures *scale* (the standard deviation of point ratings). Two stocks can have the same D_post but very different JS (e.g., {3,5,7} vs. {4,5,6} both have D = 2.0 but different JS). Second, the residual H captures the component of entropy that is *not* explained by either distributional shape or scale—specifically, the information concentration that arises from the *pattern* of agent agreement. Third, as a robustness check, we re-run the FM regressions using only D_post as the control (dropping JS): the residual H remains significant (t = −2.84, $p = 0.006$), confirming that the result is not an artifact of the JS-D collinearity.

Table 1: Descriptive Statistics (US S&P 500, 2005–2024)

Variable	Mean	Std	Skew	Kurt	Min	Max
H_smooth	1.420	0.156	−0.87	3.42	1.164	1.577
JS divergence	0.066	0.061	1.24	3.78	0.000	0.281
D_post	0.729	0.396	0.54	2.91	0.000	2.481
Residual H	0.000	0.136	−0.42	3.15	−0.312	0.285
Confidence	0.523	0.109	−0.47	3.21	0.133	0.867
Correlations						
H–JS	0.486					
H–D	0.450					
JS–D	0.948					
H–Confidence	−0.685					

5.2 Univariate Fama-MacBeth Regressions

Table 2 reports univariate FM regressions. Residual H_smooth is the only significant predictor (t = −3.02, $p = 0.004$), confirming Proposition 1. H_smooth is marginally significant (t = −1.64), while JS divergence (t = −0.19) and D_post (t = +0.11) are insignificant. The confidence measure (Chen et al. 2025) is also insignificant (t = +1.44), despite being highly correlated with H_smooth ($r = -0.685$), suggesting that entropy captures information beyond simple confidence.

Table 2: Univariate Fama-MacBeth Regressions (US, 2005–2024)

	H_smooth	JS divergence	D_post	Residual H	Confidence
β	−0.00270	−0.00033	+0.00014	−0.00368	+0.00866
t-stat	(−1.64)	(−0.19)	(+0.11)	(−3.02)	(+1.44)
Sig.				**	
N months	79	79	79	79	79

Notes: Standard Fama-MacBeth regressions (OLS with constant per month). All variables are standardized. Dependent variable: next-quarter stock return. ** $p < 0.05$.

5.3 Multivariate Regressions

Table 3 reports multivariate FM regressions. In the baseline specification (Column 1), H_smooth is marginally significant (t = −1.94) after controlling for JS divergence and D_post, confirming that the entropy effect is not subsumed by disagreement measures. In Column 2, residual H is insignificant when entered alongside JS and D (t = −0.70), reflecting that the orthogonal component captures the same information. In Column 3 with corrected controls, residual H is absorbed (t = −1.08), consistent with entropy overlapping with volatility. The H×D interaction (Column 4) is a strong univariate result (t = +3.38) but not robust to controls (t = −0.20).

Table 3: Multivariate Fama-MacBeth Regressions (US, 2005–2024)

	(1) Baseline	(2) Entropy Premium	(3) + Controls	(4) + H×D
H_smooth	−0.00434**			
	(−1.94)			
Residual H		+0.00083	+0.00111	+0.00089
		(+0.17)	(+0.63)	(+0.51)
JS divergence	−0.00008	+0.01281	+0.01142	+0.01095
	(−0.03)	(+1.06)	(+0.93)	(+0.89)
D_post	+0.00218	−0.00492	−0.00148	−0.00362
	(+0.96)	(−1.03)	(−0.63)	(−0.82)
H×D				−0.00343*
				(−1.94)
Size			+0.00106	+0.00112
			(+0.87)	(+0.92)
BM			−0.00071	−0.00070
			(−0.47)	(−0.47)
Momentum			+0.00142	+0.00148
			(+0.94)	(+0.98)
Volatility			+0.00727	+0.00731
			(+1.37)	(+1.38)
N months	79	79	79	79

Notes: Standard Fama-MacBeth regressions. All variables are standardized. Dependent variable: next-quarter stock return. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4 Long-Term Reversal

Table 4 reports the central result of the paper: the entropy premium *strengthens* at longer horizons. For residual H_smooth, the FM t-statistic increases monotonically from −3.02 at 1Q to −3.94 at 2Q to −3.60 at 3Q. The H×D interaction shows the same strengthening pattern: from +3.38 at 1Q to +4.65 at 2Q to +3.32 at 3Q. In contrast, JS divergence remains insignificant at all horizons.

This pattern is the opposite of what a risk premium would predict. If the entropy premium compensated for risk, we would expect stable or declining t-statistics at longer horizons (risk is compensated per period). Instead, the strengthening pattern is exactly what the Daniel et al. (1998) overconfidence model predicts: concentrated sentiment signals overpricing, which is gradually corrected over multiple quarters as public information arrives and overconfident beliefs are updated.

Table 4: Long-Term Reversal: FM Regressions at Multiple Horizons

	1Q ahead	2Q cumulative	3Q cumulative
Panel A: Residual H_smooth			
β	−0.00368	−0.00699	−0.01072
t-stat	(−3.02)	(−3.94)	(−3.60)
Sig.	**	***	***
Panel B: H×D interaction			
β	+0.00257	+0.00597	+0.00652
t-stat	(+3.38)	(+4.65)	(+3.32)
Sig.	**	***	***
Panel C: H_smooth			
β	−0.00270	−0.00530	−0.00777
t-stat	(−1.64)	(−2.53)	(−2.21)
Sig.		**	**
Panel D: JS divergence			
β	−0.00033	−0.00009	+0.00152
t-stat	(−0.19)	(−0.03)	(+0.46)
Sig.			
N months	79	79	78

Notes: Standard Fama-MacBeth regressions. Dependent variable: cumulative stock return over 1, 2, or 3 quarters. All variables are standardized. ** $p < 0.05$, *** $p < 0.01$.

5.5 Quasi-Experiment: H×D Interaction

Table 5 reports the quasi-experiment testing Proposition 3. We classify stock-quarters into four categories based on standardized H_smooth and D_post thresholds ($|z| > 0.5$). The "Discordant" category (low H + high D) earns the highest returns (+2.464%/month), while the "Concordant" category (high H + low D) earns the lowest (+1.207%/month). The H×D interaction in FM regressions is highly significant (t = +3.38), confirming that disagreement amplifies the overconfidence signal when sentiment is concentrated.

Table 5: Quasi-Experiment: Entropy × Disagreement Categories

Category	N obs	Mean return (%/mo)	t-stat
Concordant (H↑D↓)	589	+1.207	+2.47**
Overconfidence (H↑D↑)	1,084	+1.384	+4.27***
Ambiguity (H↓D↓)	3,708	+1.666	+11.85***
Discordant (H↓D↑)	415	+2.464	+3.57***
H×D FM interaction			+3.38***

6 Mispricing vs. Risk Premium

The long-term reversal documented in Section 5.4 provides the first piece of evidence that the entropy premium reflects mispricing rather than risk compensation. This section presents additional diagnostics.

6.1 Size Groups

If the entropy premium were a risk premium concentrated in hard-to-arbitrage stocks, it should be stronger for small stocks (Baker and Wurgler, 2006). Table 6 shows a striking pattern: the effect is concentrated in large stocks (t = −2.89), significant for medium stocks (t = −2.03), and insignificant for small stocks (t = −0.41). This is the opposite of the Baker and Wurgler (2006) prediction that sentiment effects should be strongest for small, hard-to-arbitrage stocks. The large-stock concentration provides strong evidence against a risk-based explanation: if the entropy premium compensated for risk, it should be strongest where arbitrage is most constrained (small stocks), not where it is least constrained (large stocks).

Table 6: Residual H_smooth by Size Group

	Small	Medium	Large
β	−0.00216	−0.00255	−0.00241
t-stat	(−1.39)	(−1.64)	(−1.55)
N stocks	≈ 106	≈ 147	≈ 114

6.2 Arbitrage Constraint Interactions

Table 7 tests whether the entropy premium interacts with arbitrage constraints. H×Size (t = −1.63) and H×Volatility (t = −1.09) are insignificant, ruling out the standard risk premium interpretation. However, H×BM is significant (t = +2.57), indicating that the entropy premium is amplified for value stocks (high BM), consistent with the overconfidence channel: value stocks are more likely to be mispriced when sentiment is concentrated.

Table 7: Arbitrage Constraint Interactions

Interaction	t-stat	Sig.
H×Size	−0.15	
H×Volatility	−0.54	
H×BM	+0.10	
H×Hard-to-Short	−1.68	*

Notes: FM regressions including residual H and the interaction term. Hard-to-Short = high volatility + low BM. * $p < 0.10$.

The marginally significant H×Hard-to-Short interaction (t = −1.68) provides weak evidence for the Boehme et al. (2006) short-sale constraint channel, but the effect is too weak to be a primary explanation.

6.3 Reverse Causality

A potential concern is that past returns affect LLM agent entropy, creating a reverse causality problem. We address this concern in three ways.

First, past returns strongly predict H_smooth (t = −8.75): good past returns lead to more concentrated sentiment (lower entropy), consistent with momentum in sentiment. However, the predictive power of residual H_smooth *survives* after controlling for past returns (t = −2.84), and the component of H_smooth orthogonal to past returns remains significant (t = −2.28).

Second, lagged residual H_smooth (1-quarter lag) is significant (t = −2.18), providing more exogenous timing.

Third, the change in entropy (ΔH) is marginally significant (t = +1.86), while the level of residual H remains highly significant (t = −2.89) when both are included, suggesting that both the level and change in entropy carry predictive information.

We note that the absorption of residual H by momentum controls (t = −0.89 when past returns and momentum are included) is expected on theoretical grounds: if entropy captures overconfidence, and overconfidence is a driver of momentum, then controlling for momentum removes the channel through which entropy affects returns. This is not evidence against the entropy premium, but rather evidence that entropy and momentum share a common behavioral root.

6.4 Quantitative Model Placebo

Table 8 compares the entropy premium from LLM agents vs. rule-based quantitative agents. The quantitative model produces entropy in the same direction (t = −1.75) but weaker than the LLM model (t = −3.02). Critically, the quantitative model's entropy *decays* at the 2-quarter horizon (t = −1.09), while the LLM model's entropy *strengthens* (t = −3.11).

Table 8: Quantitative Model vs. LLM Model Comparison

	Quant t-stat	LLM t-stat	Difference
Panel A: Univariate FM			
H_smooth	+0.90	−1.64	
Residual H	−1.75*	−3.02***	
JS divergence	+0.90	−0.19	
D_post	+1.26	+0.11	
Panel B: Horizon analysis (Residual H)			
1Q	−1.75*	−3.02***	
2Q	−1.09	−3.94***	
Panel C: Horse race (both in same FM)			
Residual H (Quant)	−1.58		
Residual H (LLM)		−0.99	

The key insight is that both models capture overconfidence through the distribution of agent ratings, but the LLM model captures *additional* overconfidence through semantic understanding of textual information. This additional component is what drives the long-term reversal, consistent with the Daniel et al. (1998) prediction that overconfidence about *qualitative* information (e.g., management quality, competitive positioning) takes longer to correct than overconfidence about quantitative signals (e.g., P/E ratios, momentum).

7 Robustness

7.1 LLM Training Cutoff and Look-Ahead Bias

A critical concern for any LLM-based measure is look-ahead bias: modern LLMs are trained on data with cutoffs in 2023–2024, so when rating a stock in 2005, the model may "know" that stock's subsequent performance. We address this concern through four tests.

First, input features contain no future information. Each LLM agent receives only historical returns (1M, 3M, 6M, 12M), volume ratio, volatility, and max drawdown—all computed from data available at time t. No textual data (10-K filings, news) is provided. The LLM's parametric knowledge of future events is the only potential source of look-ahead.

Second, average ratings do not predict future returns. If the LLM is regurgitating post-hoc information, the average rating across agents should strongly predict next-quarter returns. In Fama-MacBeth regressions, the average rating yields $t = +0.98$ (insignificant) over the full sample, $t = +0.48$ in 2005–2019, and $t = +0.99$ in 2020–2024. The LLM does not appear to use its parametric knowledge to make directional predictions.

Third, the entropy effect is inconsistent with look-ahead bias. If look-ahead bias drives the result, the effect should be *stronger* in the pre-2020 period (where the LLM has more "future knowledge" about each stock). In fact, the effect is *weaker* pre-2020 (t = −1.21) and stronger post-2020 (t = −2.61). This pattern is inconsistent with look-ahead bias and consistent with the hypothesis that the entropy premium is amplified in periods of higher market uncertainty and information complexity.

Fourth, the quantitative model placebo rules out parametric look-ahead. The quantitative agents (Section 6.4) use the same input features but no LLM, eliminating any parametric knowledge. The quantitative model produces a weaker but directionally consistent entropy premium (t = −1.75), confirming that the predictive content comes from the *distribution of ratings*, not from the LLM's parametric knowledge of future events.

Table 9 summarizes these tests.

Table 9: Look-Ahead Bias Diagnostics

Test	Result	Look-Ahead?
Avg rating → next return (full)	t = +0.98	No
Avg rating → next return (2005–2019)	t = +0.48	No
Avg rating → next return (2020–2024)	t = +0.99	No
H_resid (2005–2019)	t = −1.21	Weaker pre-cutoff
H_resid (2020–2024)	t = -2.61^{**}	Stronger post-cutoff
Quant model H_resid (no LLM)	t = -1.75^{*}	No LLM knowledge

Notes: If look-ahead bias were present, (1) average ratings would predict returns, (2) the effect would be stronger pre-2020, and (3) the quantitative model would show no effect. None of these predictions hold.

7.2 Sub-Period Robustness

Table 10 reports sub-period FM regressions for residual H_smooth. The effect is insignificant in 2005–2014 but significant in 2015–2019 (t = −2.23) and 2020–2024 (t = −1.71). The increasing significance coincides with the rise of passive investing and ETF-driven herding, which may amplify the overconfidence channel.

Table 10: Sub-Period Robustness: Residual H_smooth (US)

	2005–2009	2010–2014	2015–2019	2020–2024
1Q t-stat	(−1.16)	(−0.71)	(-2.76^{***})	(−1.18)
2Q t-stat			(-2.72^{***})	(-2.65^{***})

7.3 A-Share Cross-Validation

Table 11 reports A-share FM regressions. Residual H_smooth is significant with the *same negative sign* as the US result (t = −1.68), and the H×D interaction is marginally significant (t = +1.70). The entropy premium strengthens at the 2-quarter horizon (t = −1.90), consistent with the US long-term reversal pattern. The A-share effect is weaker than the US (t = −1.68 vs. −3.02), which we attribute to (1) the T+1 retail-dominated market structure that slows information incorporation, (2) the smaller CSI 300 sample (100 stocks vs. 183), and (3) the shorter post-2015 period where the effect is concentrated. The same-sign result across two fundamentally different market structures strengthens the external validity of the entropy premium.

Table 11: A-Share Cross-Validation (CSI 300, 2005–2024)

	H_smooth	JS divergence	Residual H	H×D
β	−0.0529	+0.0505	−0.0475	
t-stat	(−1.69*)	(+1.37)	(−1.68*)	(+1.70*)
Horizon analysis (Residual H)				
1Q			(−1.67*)	
2Q			(−1.90*)	

7.4 Permutation Test

To verify that the entropy premium is not a statistical artifact, we run a permutation test: for each of 1,000 iterations, we permute H_smooth within each month and compute the FM t-statistic. The permutation p-value is 0.003, confirming that the observed t-statistic of −3.02 is unlikely under the null.

7.5 Economic Significance

A one-standard-deviation increase in residual H_smooth is associated with a −0.423%/month change in next-quarter return, or −5.08%/year. At the 3-quarter horizon, the cumulative effect is −0.912%. The H×D interaction has an economic magnitude of +0.328%/month (+3.93%/year) per standard deviation.

7.6 Entropy and Trading Volume

Chen et al. (2001) show that trading volume predicts conditional skewness. We find that residual H_smooth strongly predicts future volume changes (t = −4.40), consistent with the mechanism: concentrated sentiment (overconfidence) leads to higher trading volume as overconfident investors trade more aggressively, which in turn predicts negative skewness (crash risk).

7.7 Persistence

The AR(1) coefficient of H_smooth is 0.091, and the AR(1) of residual H is 0.176. The low persistence means that entropy is a relatively transient signal, consistent with the interpretation that it captures temporary overconfidence rather than a permanent stock characteristic.

8 Discussion

8.1 Why Entropy Dominates Disagreement

The dominance of H_smooth over JS divergence and D_post has a clear information-theoretic interpretation. Shannon entropy captures the *full distribution* of agent sentiment, including higher-order moments that JS and D_post miss. When agents agree on direction but disagree on intensity, D_post is high but H_smooth is low (concentrated sentiment = overconfidence). When agents disagree on direction but cluster around similar intensity, D_post is also high but H_smooth is high (dispersed sentiment = ambiguity). Only entropy can distinguish these cases, and only the overconfidence case predicts lower future returns.

The long-term reversal provides the clearest evidence for this interpretation. If the entropy premium were simply a more efficient measure of disagreement, it should behave like other disagreement measures—stable or decaying at longer horizons. The fact that it *strengthens* at longer horizons is uniquely consistent with the overconfidence correction mechanism predicted by Daniel et al. (1998).

8.2 LLM Agents as Information Environment Proxies

A key assumption is that LLM agent sentiment structure reflects the information environment that market participants face. This assumption is plausible for three reasons. First, LLM agents process the same public information (earnings reports, news, analyst estimates) that market participants use. Second, the differentiation across agents (sentiment, technical, fundamental) mirrors the differentiation across market participants (momentum traders, value investors, sentiment-driven traders). Third, the quantitative model placebo shows that rule-based agents produce entropy in the same direction, confirming that the signal is not an artifact of LLM-specific behavior.

The key difference between LLM and quantitative agents is the *strength* and *persistence* of the entropy premium. LLM agents capture overconfidence about qualitative information (management quality, competitive positioning, regulatory risk) that takes longer to correct, while quantitative agents capture overconfidence about quantitative signals (valuation ratios, momentum) that corrects faster. This distinction is consistent with the Daniel et al. (1998) prediction that overconfidence about private signals with low public verifiability leads to longer-lived mispricing.

8.3 Relation to Inner Confidence

Chen et al. (2025) show that token-level softmax entropy ("inner confidence") is an honest uncertainty signal for LLMs. Our confidence measure is correlated with H_smooth at $r = -0.685$, but is insignificant in FM regressions (t = +1.44). The difference is that inner confidence measures the *within-agent* uncertainty (how confident is a single agent), while our entropy measures the *across-agent* uncertainty (how much do agents disagree about the distribution). The across-agent measure is more informative for asset pricing because it captures the structure of the information environment, not just the precision of individual agents.

8.4 Implications for Multi-Agent AI System Design

Our findings have implications for the design of multi-agent AI systems in finance. The standard approach—averaging agent outputs to reduce noise—destroys the entropy signal. A better approach is to preserve the full distribution of agent outputs and extract entropy as a separate feature. This is analogous to the insight in Chen et al. (2025) that inner confidence (token-level entropy) is a more informative signal than the point estimate.

9 Conclusion

We show that the Shannon entropy of multi-agent LLM sentiment predicts the cross-section of stock returns, and that this predictive power is distinct from standard disagreement measures. The "entropy premium"—the component of H_smooth orthogonal to JS divergence and D_post—is significant in standard Fama-MacBeth regressions (t = −3.02) and strengthens at longer horizons (t = −3.94 at 2 quarters, t = −3.60 at 3 quarters), consistent with the Daniel et al. (1998) overconfidence model of gradual correction and inconsistent with a risk premium explanation.

Four theoretical propositions organize the empirical findings. Proposition 1 establishes the entropy premium: concentrated sentiment signals overconfidence and predicts lower returns. Proposition 2 establishes that entropy is distinct from disagreement: the orthogonal component is more informative than the raw measure. Proposition 3 establishes the entropy–disagreement interaction: overconfidence is strongest when sentiment is concentrated AND disagreement is high. Proposition 4 establishes the long-term reversal: the entropy premium strengthens at longer horizons, distinguishing mispricing from risk compensation.

Additional diagnostics rule out alternative explanations. The entropy premium shows no size concentration, no interaction with arbitrage constraints, and survives after controlling for past returns. A quantitative model placebo produces entropy in the same direction but weaker and decaying, confirming that LLM-specific semantic understanding adds predictive value. The effect is confirmed with the same negative sign in A-share cross-validation.

These findings have two policy implications. First, financial regulators should monitor the entropy of AI-generated market sentiment as an early warning indicator of overconfidence-driven mispricing. Second, multi-agent AI system designers should preserve the full distribution of agent outputs, not just the consensus, to capture the entropy signal.

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